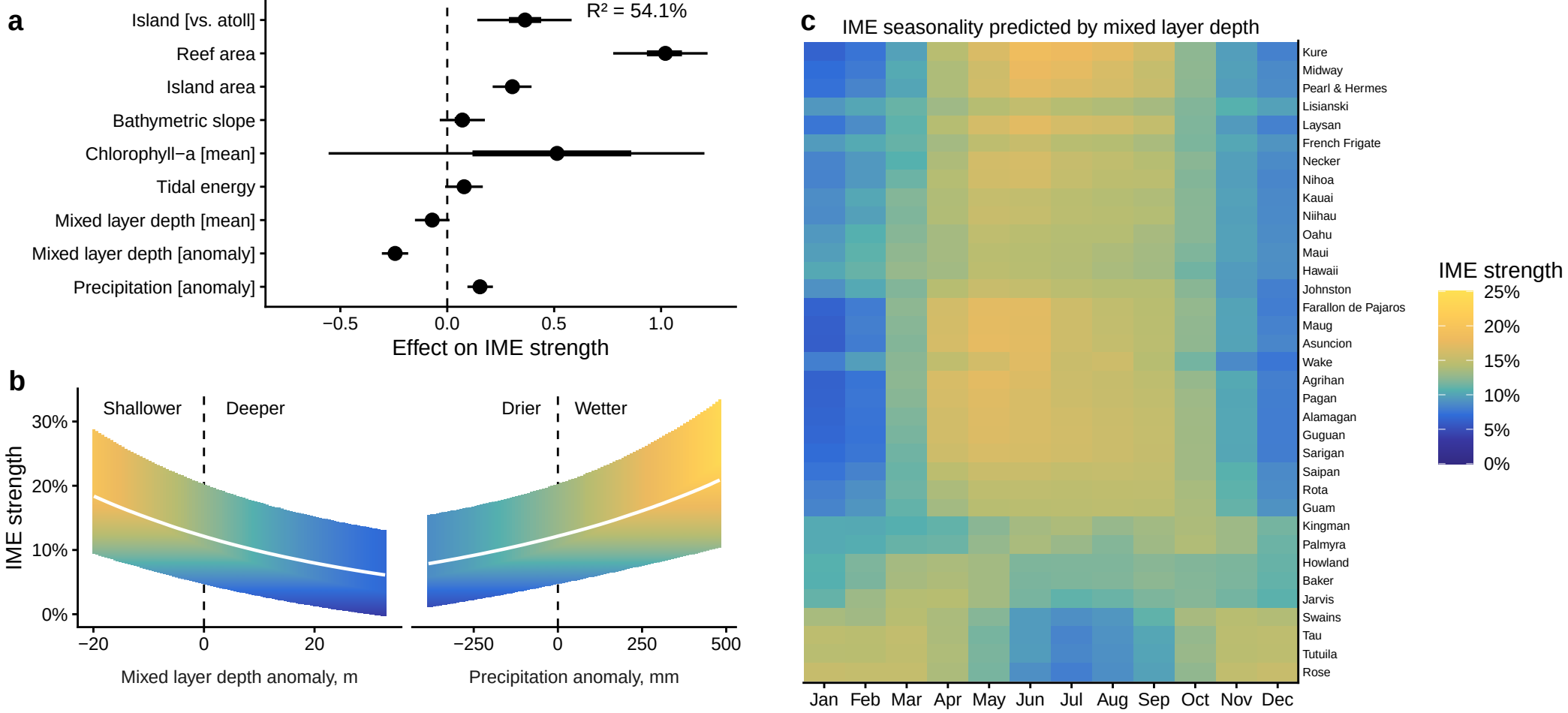
a
R² = 54.1%
Island [vs. atoll]
Reef area
Island area
Bathymetric slope
Chlorophyll−a [mean]
Tidal energy
Mixed layer depth [mean]
Mixed layer depth [anomaly]
Precipitation [anomaly]
−0.5
0.0
0.5
1.0
Effect on IME strength
b
Shallower
Deeper
Drier
Wetter
IME strength
30%
20%
10%
0%
−20
0
20
−250
0
250
500
Mixed layer depth anomaly, m
Precipitation anomaly, mm
c
IME seasonality predicted by mixed layer depth
Kure
Midway
Pearl & Hermes
Lisianski
Laysan
French Frigate
Necker
Nihoa
Kauai
Niihau
Oahu
Maui
Hawaii
Johnston
Farallon de Pajaros
Maug
Asuncion
Wake
Agrihan
Pagan
Alamagan
Guguan
Sarigan
Saipan
Rota
Guam
Kingman
Palmyra
Howland
Baker
Jarvis
Swains
Tau
Tutuila
Rose
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
IME strength
25%
20%
15%
10%
5%
0%